

ACCURATE OBJECT MATCHING FOR UAV IMAGERY USING MULTI-SCALE BEST-BUDDIES SIMILARITY

Xu Lei^{1,2,3}, Jinwang Wang^{1,2}, Kaimin Fu^{1,2}, Huai Yu^{1,2}, Wen Yang^{1,2}

¹School of Electronic Information, Wuhan University, Wuhan 430072, China

²Wuhan University DJI Joint Laboratory of Unmanned Aerial Vehicles, Wuhan 430072, China

³School of Software Engineering, Huazhong Univ. of Sci. & Technol., Wuhan 430074, China

ABSTRACT

This paper presents a multi-scale Best-Buddies Similarity (BBS) method for matching objects in the UAV imagery. More precisely, we first extract proposed regions from the target image by Selective Search, then take a number of proposed regions with the highest confidence scores as templates and finally apply templates to the original BBS method and select the coordinate of the region with the highest overlap rate as the exact position of the query object in the target image. Experimental results on real data show the effectiveness of the proposed method.

Index Terms— multi-scale, object matching, best-buddies similarity, selective search, unmanned aerial vehicle

1. INTRODUCTION

Object matching has been increasingly exploited in such applications as face recognition and medical imaging. Our interest is in object matching in the unmanned aerial vehicle (UAV) imagery. A typical use case is in intelligent surveillance, where the UAV utilizes object matching to recognize the specific object in various states. Another application scenario is in UAV tracking, where object matching is applied to locate the specific object under different environmental conditions. In both above applications of UAV, a robust object matching method is quite necessary. For this problem, we consider utilizing similarity measures in the object matching algorithm. Traditional similarity measures include Sum of Squared Differences (SSD) and Normalized Cross-Correlation (NCC) [1]. The main idea of SSD is to calculate pixel's differences between two images. More specifically, it squares the sum of subtraction between two images' pixels. NCC determines the matching point between template and image by searching the location of maximum value in the image matrices. Compare with SSD and NCC, Best Buddies Similarity (BBS) [2] proposed by Oron et al. is more robust to cluttered background, occlusions and non-rigid deformation, which makes it an ideal similarity measure for UAV applications. Specifically, BBS is a similarity measure that maps both the template and the patch (generated from the target image by

sliding window) into the point set. For the point in the point set, only a pair of the points that are mutual nearest neighbors can be called the Best-Buddies Pairs. Then, the BBS is defined as the total number of the Best-Buddies Pairs. However, for the reason that the window size is fixed, the original BBS method cannot account for scale changes. This limits the application of the BBS method in the UAV imagery. Thus, finding a multi-scale BBS method utilized for UAV imagery is our main concern.

In order to address this concern, we introduce an approach using object proposal to detect the actual scale of object in the target image. Object proposal is a methodology to guide the search of objects. It can be mainly divided into two categories. One category of object proposal is grouping methods. The other category of object proposal is window scoring methods. Representative methods of grouping are selective search [3] and randomized prim's algorithm [4]. Selective search generates proposed regions by greedily merging super pixels. Randomized prim's algorithm is in somehow more efficient by introducing a randomized super-pixel merging process. Representative methods of window scoring are bing [5] and edge boxes [6]. Without any prior knowledge, the edge boxes method is realized by estimating the boundary of objects using geometric method. Bing is a very fast method by using a simple linear classifier. Among all the object proposal methods, selective search has the relatively best performance. Hence, we apply selective search to propose regions in this work. In contrast with the original BBS, our work introduces the idea of object proposal to solve the multi-scale problem in the scale space. To our knowledge, some recent works close to ours include [7], which introduces the idea of clustering and random sampling for Best-Buddies tracking. However, this method is under the framework of particle filtering, which is very different from ours.

The remainder of the paper is organized as follows. In Section 2, we illustrate three major stages of the proposed method and explain the idea of using the object proposal in BBS. In section 3, experimental results of the proposed method for matching both rigid and non-rigid objects, are reported. Conclusion is drawn in Section 4.

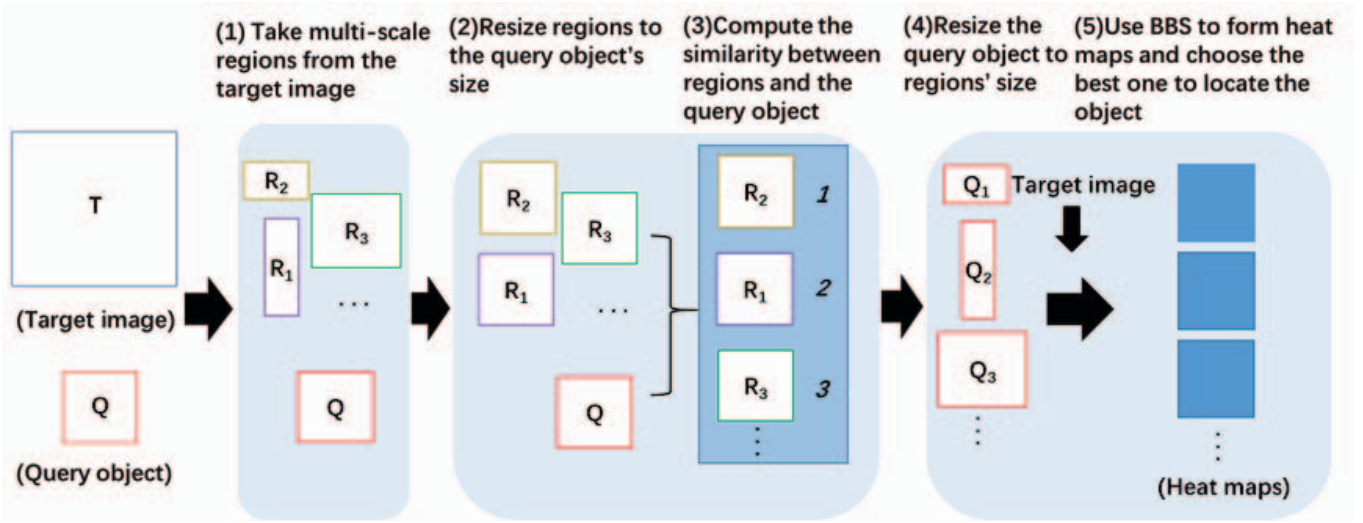


Fig.1 The flowchart of BBS using object proposal. Several multi-scale regions are extracted from target image in (1), resized to the size of the query object in (2) and compared with the query object in (3). After comparing, we calculate confidence scores for each region. According to confidence scores, we take a certain number of regions as templates. Then the query object is resized to the size of the selected template in (4) to infer the observation likelihood map of the target image in (5). Later, the observation likelihood map satisfied certain rules is utilized to locate the query object. More details will be provided next.

2. PROPOSED METHODOLOGY

In this section, we aim to present the idea of our methodology. Our processing workflow mainly consists of three major stages (in blue boxes shown in the Fig.1) including the extraction of proposed multi-scale regions from the query image, the sorting of proposed regions based on BBS method, the localization of the query object in the target image using BBS method. We now describe each of them in details.

2.1. Proposed Regions Using Object Proposal Method

Object proposal is a technique for distinguishing candidate objects from the background. Unlike exhaustively sliding along the whole image, object proposal efficiently output a set of proposed regions contained detected objects. Therefore, it has been widely applied to detect scales and positions of objects proposed from the image [8]. In this work, we choose selective search as the object proposal method to generate multi-scale regions. Selective search iteratively groups super-pixels to generate proposed regions. In addition, selective search possesses the property of capturing multi-scale objects without acquiring prior knowledge. This property helps to estimate the actual scale of the query object in the target image for later analysis. In the end, the basic algorithm of selective search is described as follow [3]:

Algorithm 1: Hierarchical grouping algorithm

Input: UAV image

Output: scale and position of proposed regions

Segment image into initial regions $R = \{r_1, \dots, r_n\}$ using [9]

Initialize similarity set $S = \emptyset$

for every neighboring region pair (r_i, r_j) **do**

 Calculate similarity $s(r_i, r_j)$

$S = S \cup s(r_i, r_j)$

end

while $S \neq \emptyset$ **do**

 Get the highest similarity $s(r_i, r_j) = \max(S)$

 Merge corresponding regions $r_m = r_i \cup r_j$

 Remove similarities regarding r_i : $S = S \setminus s(r_i, r_i)$

 Remove similarities regarding r_j : $S = S \setminus s(r_j, r_j)$

 Calculate similarity set S_m between r_m and its neighbors

$S = S \cup S_m$

$R = R \cup r_m$

end

Extract object location and scale from $\{R_1, \dots, R_n\}$ in R

2.2. Regions Sorting Based on BBS

In the first stage, various regions in different scales have been drawn from the target image. However, not all proposed regions might contain the detected object we need. Thus, the confidence score is employed to assess the probability that proposed regions contain the target object. Confidence score is a certain similarity measure. Here, we adopt BBS to evaluate the similarity between the query object and proposed regions. BBS can be written as [2]:

$$bb(p_i, q_j, P, Q) = \begin{cases} 1 & NN(p_i, Q) = q_j \wedge NN(q_j, P) = p_i \\ 0 & \text{others} \end{cases} \quad (1)$$

$$BBS(P, Q) = \frac{1}{\min\{M, N\}} \sum_{i=1}^N \sum_{j=1}^M bb(p_i, q_j, P, Q) \quad (2)$$

where $NN(p_i, Q) = \arg \min_{q \in Q} d(p_i, q)$, $P = \{p_i\}_{i=1}^N$, $Q = \{q_i\}_{i=1}^M$
 $d(p_i, q_j)$ is a certain distance measure. Specifically, in this stage,

$$d(p_i, q_j) = \|p_i^{(D)} - q_j^{(D)}\|_2^2 \quad (3)$$

where superscript D denotes a dense-SIFT descriptor.

The final process is summarized in algorithm 2:

Algorithm 2: Region sequencing algorithm

Input: proposed regions

Output: confidence score of proposed regions, sequence of proposed regions after sorting
 divide Q into non-overlap cells (k*k pixels)
 extract a dense-SIFT descriptor from each cell to form the descriptor set M

for each *proposed region* R_i **do**

 resize R_i to the size of query object Q

 divide R_i into non-overlap cells (k*k pixels)

 extract a dense-SIFT descriptor from each cell to form the descriptor set N

 confidence score = $BBS(M, N)$

end

sort proposed regions by the confidence score

2.3. Object Localization Using BBS

In the second stage, proposed regions have been sorted by confidence score. Then, we can sequentially get the overlap rate of each proposed region by employing algorithm 3. Nevertheless, in this way, the computational cost is very huge. Hence, we just take a number of regions with the highest confidence scores as templates in this work. The details of utilizing these templates are summarized in algorithm 3. Specifically, BBS used for algorithm 3 utilizes the same distance measure as in the original BBS work, which is defined as:

$$d(p_i, q_j) = \|p_i^{(A)} - q_j^{(A)}\|_2^2 + \lambda \|p_i^{(L)} - q_j^{(L)}\|_2^2 \quad (4)$$

where superscript A denotes a points appearance and superscript L denotes a points location.

Algorithm 3: Positioning algorithm

Input: sorted regions, UAV image

Output: precision position of the query object in the target image

take a number of regions with the highest confidence scores as templates

for each *template* R_i **do**

 resize the query image Q to the size of R_i

 create patches (the same size as Q) from the target

image using the sliding window

 divide R_i into non-overlap cells (k*k pixels)

 extract appearance and location information from each cell to form the set E

for each *patch* **do**

 divide patch into non-overlap cells (k*k pixels)

 extract appearance and location information from each cell to form the set F

 patch score = $BBS(E, F)$

end

 incorporate patch scores into the heat map

 find the position of the global maxima in the heat map as the detected region's position

 calculate the overlap rate between R_i and the

$$\text{detected region } D_i: \text{Overlap} = \frac{|R_i \cap D_i|}{|R_i \cup D_i|}$$

end

output the position of the detected region with the highest overlap rate as the final result

3. EXPERIMENTAL RESULTS

Two kinds of experimental object are tested in this section. One is the rigid object. The other is the non-rigid object. The image of our experiment was generated from UAV videos and its size is 640×480 pixels. The video was acquired over the road of Wuhan University by PHANTOM 4 pro. Query objects were manually chosen and labeled in green boxes in Fig.2. Detected results of our method and the original BBS method are labeled in blue boxes and red boxes in Fig.2 respectively. The analysis of these experimental results is as follows.

3.1. Rigid Object

The rigid object is the object on which the distance between two points never changes wherever it moves to. Here, we use the electric bike and the car as test objects. Results of electric bike are shown in the first row of Fig.2. Results of car are shown in the second row of Fig.2.

In the experiment of electric bike, both Fig.2(a) and Fig.2(b) were acquired at the height of twenty meters. By using the method of selective search, we took 338 proposed regions from Fig.2(b). After sorting them, we selected ten regions with the highest confidence scores as templates. Through applying these templates to the method proposed in section 2.3, we got the precision position of the query object labeled in the blue box in Fig.2(c). Comparing with the result presented in Fig.2(d), our method can match the electric bike well when its scale becomes much smaller.

In the experiment of car, we took 308 proposed regions from Fig.2(f) by using the method of Selective Search. After sorting them, ten regions with the highest confidence scores

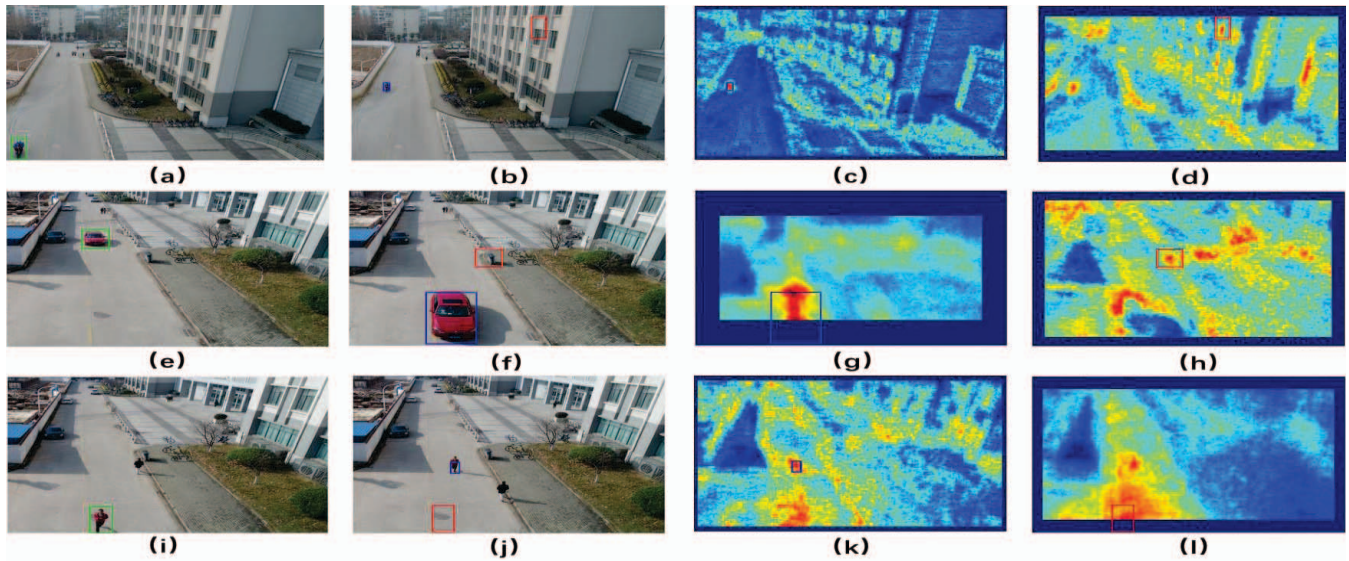


Fig.2 Results on real data. The first column is consisted of images contained query images. The second column includes the corresponding target images. The third and fourth columns are consisted of likelihood maps generated by our method and original BBS method respectively.

were selected as templates. Through applying these templates to the method proposed in section 2.3, we got the precision position of the query object labeled in the blue box in Fig.2(g). Comparing with the result presented in Fig.2(h), our method can match the car well when the scale of the car becomes much larger.

In these two experiments, the original BBS method cannot work well for scale changes of rigid objects. But our method can match them well when the scale is changed. That means our method is efficient for rigid objects.

3.1. Non-rigid Object

The non-rigid object is the object on which the distance between two points will change when it moves. Here, we utilize the walking person as the test object. The original images and detection results are shown in the third row of Fig.2.

In this experiment, 324 proposed regions were taken from Fig.2(j) by using selective search. After applying the method proposed in section 2.2, ten regions with the highest confidence scores were chosen as templates. Then, these templates were used for the method proposed in section 2.3 to locate the query object in the target image. After that, we got the location of the query object labeled in the blue boxes in Fig.2(k). Comparing with the result in Fig.2(I), our method is more robust for scale changes of the non-rigid object. That is to say our method achieves superior performance in the scale space and more suitable for object matching.

4. CONCLUSION

In this paper, we have presented a multi-scale BBS approach by utilizing selective search method. Three main

steps are presented to illustrate the processing workflow. Two kinds of experimental object are used to demonstrate the effectiveness of the proposed method. One limitation of our algorithm is that the region sorting method we used is not accurate enough and cannot ensure the query object always existing in the template list. In light of this, our future work will focus on finding a precise region sorting method to achieve more stable results.

REFERENCES

- [1] M. B. Hisham, S. N. Yaakob, R. A. A Raof, A.B A. Nazren, and N.M. Wafi Embedded, "Template Matching using Sum of Squared Difference and Normalized Cross Correlation," in *proc. SCOREd*, 2015, pp. 100-104
- [2] T. Dekel, S. Oron, S. Avidan, M. Rubinstein, and W. Freeman, "Best Buddies similarity for robust template matching," in *proc. CVPR*, 2015, pp. 2021 – 2029.
- [3] J. Uijlings, K. van de Sande, T. Gevers, and A. Smeulders, "Selective search for object recognition," *IJCV*, 2013, pp. 154-171.
- [4] S. Manén, M. Guillaumin, and L. Van Gool, "Prime object proposals with randomized prim's algorithm," in *proc. ICCV*, 2013, pp. 2536-2543.
- [5] M.-M. Cheng, Z. Zhang, W.-Y. Lin, and P. H. S. Torr, "BING: Binarized normed gradients for objectness estimation at 300fps," in *proc. CVPR*, 2014, pp. 3286-3293.
- [6] C. Zitnick and P. Dollár, "Edge boxes: Locating object proposals from edges," in *proc. ECCV*, 2014, pp. 391-405.
- [7] S. Oron, D. Suhanov, and S. Avidan, "Best-Buddies Tracking," *arXiv:1611.00148*, 2016.
- [8] J. Hosang, R. Benenson, P. Dollár, B. Schiele, et al, "What Makes for Effective Detection Proposals?" *IEEE PAMI*, 2016, pp. 814-830.
- [9] P. F. Felzenszwalb and D. P. Huttenlocher, "Efficient Graph Based Image Segmentation," *IJCV*, 2004, pp. 167–181.